

Lecture 23

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CYCLIC EXTENSIONS

THM: K/F GALOIS

Assume $\text{Gal}(K/F)$ CYCLIC WITH σ GENERATOR

Let $x \in K$

$$N_{K/F}(x) = 1 \iff \exists y \in K \text{ with } x = \frac{y}{\sigma(y)}$$

$$\begin{array}{ccc} K & \xrightarrow{a \mapsto a/\sigma(a)} & K^x \\ \downarrow & & \downarrow \\ K^x & \xrightarrow{\text{GROUP HOMOS}} & F^x \end{array}$$

$$n = |\text{Gal}(K/F)| \quad N_{K/F}(x) = x \sigma(x) \cdots \sigma^{n-1}(x)$$

$$\frac{N(a)}{N(\sigma(a))} = 1$$

$$1, \sigma, \dots, \sigma^{n-1}: K \rightarrow K$$

LEMMA: G GROUP, K FIELD

$$\chi_1, \dots, \chi_n: G \rightarrow K^x \text{ GROUP HOMOS \& } \chi_i \neq \chi_j$$

$$\text{IF } a_1 \chi_1 + \dots + a_n \chi_n: G \rightarrow K \text{ VANISHES,} \\ a_1 = \dots = a_n = 0.$$

PF: INDUCTION ON n .

OK FOR $n=1$.

Assume $n \geq 2$

$$\chi_1 = \chi_2 \Rightarrow \exists g \in G \text{ st } \chi_1(g) \neq \chi_2(g)$$

$$a_1 \chi_1 + \dots + a_n \chi_n = 0 \quad h \in G \quad \chi_i(gh) = \chi_i(g) \chi_i(h)$$

$$\Rightarrow a_1 \chi_1(g) \chi_1(h) + a_2 \chi_2(g) \chi_2(h) + \dots + a_n \chi_n(g) \chi_n(h) = 0$$

$$a_1 \chi_1(h) + a_2 \frac{\chi_2(g)}{\chi_1(g)} \chi_2(h) + \dots + a_n \frac{\chi_n(g)}{\chi_1(g)} \chi_n(h) = 0$$

$$\text{ALSO } a_1 \chi_1(h) + \dots + a_n \chi_n(h) = 0 \quad \text{so}$$

$$\underbrace{a_2 \left(1 - \frac{\chi_2(g)}{\chi_1(g)}\right)}_{a_2'} \chi_2(h) + \dots + \underbrace{a_n \left(1 - \frac{\chi_n(g)}{\chi_1(g)}\right)}_{a_n'} \chi_n(h) = 0$$

$$a_2' \chi_2(h) + \dots + a_n' \chi_n(h) = 0$$

$$\text{BY INDUCTION } a_2' = \dots = a_n' = 0, \text{ so}$$

$$a_2 \left(1 - \frac{\chi_2(g)}{\chi_1(g)}\right) = 0, \quad a_2 (\chi_1(g) - \chi_2(g)) = 0 \\ \Rightarrow a_2 = 0.$$

$$\text{INDUCTION: } a_1 \chi_1 + a_3 \chi_3 + \dots + a_n \chi_n = 0$$

$$\Rightarrow a_1 = \dots = a_n = 0. \quad \square$$

PF (THM)

$$1, \sigma, \dots, \sigma^{n-1}: K^x \rightarrow K^x \text{ DISTINCT}$$

$$\text{LET } a_1 = 1, \quad a_2 = x, \quad a_3 = x\sigma(x), \dots, a_{n-1} = x\sigma(x) \cdots \sigma^{n-2}(x)$$

$$a_1 \text{Id} + a_2 \sigma + \dots + a_n \sigma^{n-1}: K^x \rightarrow K^x \quad \text{NON-ZERO BY LEMMA}$$

$$\text{SO } \exists z \in K^x \text{ s.t.}$$

$$u = z + x\sigma(z) + x\sigma(x)\sigma^2(z) + \dots + x\sigma(x) \cdots \sigma^{n-2}(z) \neq 0$$

$$\text{Claim: } x \cdot \sigma(u) = u$$

$$x\sigma(u) = x\sigma(z) + x\sigma(x)\sigma^2(z) + \dots + \underbrace{x \cdots \sigma^{n-2}(z)}_1 = u.$$

$\square$